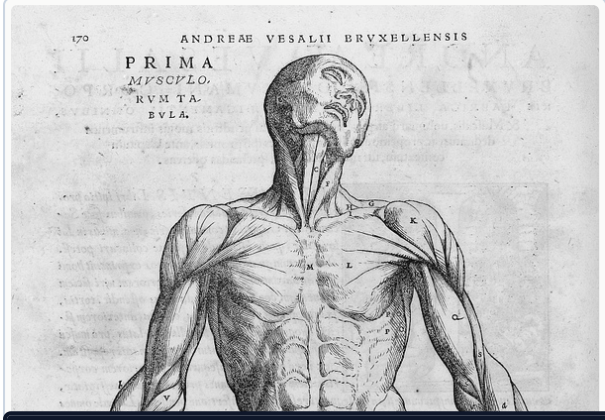
PEARSON EDEXCEL GCSE (9–1) HISTORY • PAPER 1 OPTION 11

THE MEDICAL RENAISSANCE IN BRITAIN (C1500–C1700)

Humanism, Direct Dissection & The Mechanical Revolution



HISTORICAL OVERVIEW • C1500–C1700

The collapse of feudalism, the printing press, and Renaissance humanism dismantled the Church's monopoly over medical learning. Scholars began questioning ancient texts, insisting that knowledge must rest on direct anatomical dissection and observable evidence. In 1543, Andreas Vesalius published De Humani Corporis Fabrica, proving Galen had committed over two hundred anatomical errors.

From William Harvey proving the mechanical circulation of blood in 1628 to Thomas Sydenham championing clinical bedside observation, the Scientific Revolution transformed medicine into an empirical science. Yet as the 1665 Great Plague demonstrated, revolutionary breakthroughs in anatomical understanding could not immediately overcome the ancient challenge of discovering effective medical cures.

SPECIFICATION & ENQUIRY DIRECTORY

Pearson Edexcel GCSE (9–1) History

ENQUIRY 01 pp. 3–4

The New Spirit of Enquiry: Humanism & The Royal Society

KEY QUESTION: *"How did printing and empirical science challenge 1,500 years of orthodoxy?"*

- **Renaissance Humanism:** Rediscovery of original classical Greek texts encouraged scholars to question medieval translations and observe nature directly.
- **The Printing Press (c1440):** Gutenberg's moveable metal type allowed rapid, exact copying of medical diagrams, completely bypassing Church scriptoria.
- **The Royal Society (1660):** Founded in London under royal charter with the motto *Nullius in Verba* ("Take nobody's word for it"), publishing scientific papers.

ENQUIRY 02 pp. 5–6

Thomas Sydenham & The Art of Bedside Observation (1676)

KEY QUESTION: *"How did Sydenham's focus on direct observation challenge traditional practice?"*

- **The English Hippocrates:** Refused to rely on classical textbooks; observed patients closely at bedside and took meticulous daily clinical notes.
- **Classification of Diseases:** Argued diseases were distinct species that could be classified like plants; diagnosed scarlet fever as distinct from measles.
- **Empirical Prescriptions:** Prescribed cinchona bark (quinine) for malaria, iron for anaemia, and fresh air/light blankets for smallpox fevers.

ENQUIRY 03 pp. 7–8

Andreas Vesalius & The Anatomical Revolution (1543)

KEY QUESTION: *"Why did Andreas Vesalius succeed in overturning Galenic authority?"*

- **Direct Human Dissection:** Professor of Surgery at Padua who conducted human dissections himself rather than delegating cutting to barber-surgeons.
- **De Humani Corporis Fabrica (1543):** Landmark text with accurate anatomical plates by Titian's workshop, correcting over 200 of Galen's anatomical errors.
- **Overturning Dogma:** Proved Galen dissected animals; proved the human lower jaw was one single bone and the cardiac septum had no invisible pores.

ENQUIRY 04 pp. 9–10

William Harvey & The Circulation of the Blood (1628)

KEY QUESTION: *"Why did Harvey's discovery of circulation meet fierce medical resistance?"*

- **Calculated Mechanical Output:** Proved mathematically that the liver could not produce the immense volume of blood pumped through the heart each hour.
- **One-Way Valves & Dissection:** Dissected cold-blooded animals; used arm tourniquets to prove veins have one-way valves pumping blood to the heart.
- **De Motu Cordis (1628):** Disproved Galen's theory that blood was consumed by tissues; took over 30 years for his circulation theory to be accepted.

ENQUIRY 05 pp. 11–12

Continuity in Treatment & The Great Plague of London (1665)

KEY QUESTION: *"Why did the Great Plague reveal the severe limits of Renaissance progress?"*

- **Urban Catastrophe:** Struck London in summer 1665, killing over 100,000 people (nearly 20% of the city's population) within seven months.
- **Persistent Superstitions:** Quack doctors sold useless plague waters; people smoked tobacco to ward off miasma and bled patients, continuing medieval treatments.
- **Stricter Civic Quarantine:** Mayoral watchmen locked infected houses for 28 days with red crosses ("Lord Have Mercy Upon Us"); mass burial pits outside city walls.

THEMATIC MATRIX pp. 2–11

Thematic Matrix: Longitudinal Factors & Synthesis

KEY QUESTION: *"How far did anatomical revolution translate into medical treatment?"*

- **Science & Technology:** The printing press, microscopes, and Royal Society enabled rapid international dissemination of empirical discoveries.
- **The Gap Between Anatomy and Cure:** Despite Vesalius and Harvey transforming anatomy and physiology, everyday bedside treatments remained medieval.
- **Weak Royal Intervention:** Charles II provided royal patronage to science but national government still provided zero public health funding.

SPECIFICATION PAPER	SECTION B QUESTION ARCHITECTURE	ASSESSMENT WEIGHTING
Paper 1 Option 11 • Thematic Study	Q3: Similarity/Diff [4m] • Q4: Causation [12m] • Q5/Q6: Essay [	36 Raw Marks + 4 SPaG • 70% of Paper 1 T

ENQUIRY 01 • PART I

PEARSON EDEXCEL GCSE (9–1) HISTORY • PAPER 1 (1HI0/11)

The New Spirit of Enquiry: Humanism, The Printing Press & The Royal Society (c1500–c1700)

KEY ENQUIRY QUESTION *How did the invention of printing and the scientific method challenge 1,500 years of medical orthodoxy?*

[1.1] The Printing Press and the Information Revolution (c1440–c1500)

"Why did printing break the clerical monopoly on medical learning?"

Around 1440 in Mainz, German inventor Johannes Gutenberg developed the revolutionary **Gutenberg printing press**, sparking an intellectual information revolution across Renaissance Europe. By casting reusable individual letters from durable **movable metal type**, Gutenberg mechanized book manufacturing, decisively breaking the centuries-old clerical monopoly on written manuscript knowledge previously guarded by Catholic monastic scriptoria throughout European Christendom.

When William Caxton established England's first commercial press at Westminster in 1476, **rapid book mass-production** transformed medical scholarship. Textbooks could now be printed accurately in identical editions, eliminating transcription errors made by exhausted monk copyists. Scientific ideas, anatomical diagrams, and empirical treatises circulated swiftly across European borders, allowing researchers to study, debate, and verify medical breakthroughs simultaneously in print.

[1.2] Renaissance Humanism: Questioning Ancient Authority

"How did humanism inspire physicians to challenge Galen?"

Beginning in fifteenth-century Italy, the revolutionary movement of **Humanism scholarly inquiry** transformed European thought by encouraging direct intellectual curiosity and independent critical thinking. Humanist scholars rejected blind medieval deference to Church dogma, seeking out forgotten classical manuscripts scattered across monastic libraries following the sudden fall of Constantinople in 1453. They sought to understand the natural world through human observation rather than divine revelation.

Medical humanists studied **original Greek translations** of Hippocrates and Galen, discovering that centuries of careless medieval Latin translations had distorted classical teachings. More importantly, humanism revived the ancient spirit of empirical inquiry, **challenging Galenic dogma** by urging physicians to verify anatomical descriptions through direct human dissection rather than accepting ancient authority without question. This critical mindset provided the intellectual catalyst for early modern anatomical science.

ENQUIRY 01 • PART II

TURNING POINTS & HISTORICAL CONSEQUENCES

Nullius in Verba & The Royal Society (1660)

DECISIVE HISTORICAL MECHANISM Transformed medical enquiry from passive philosophical debate into active empirical investigation.

[1.3] The Royal Society and the Spirit of Nullius in Verba (1660)

"Why was experimental observation central to the Royal Society?"

In November 1660, a distinguished gathering of natural philosophers—including architect Christopher Wren, chemist Robert Boyle, and John Wilkins—founded a new scientific academy at Gresham College in London. This landmark assembly became the prestigious **Royal Society founded (1660)**, established to promote natural knowledge through collaborative laboratory experimentation and rational debate rather than theological speculation. The society rapidly attracted Britain's leading scientific minds.

In 1662, the institution received an official **royal charter Charles II**, granting royal patronage and intellectual freedom. The Society adopted the famous **'Nullius in Verba' motto** ('Take nobody's word for it'), boldly declaring that scientific truth could never rely on ancient authority, but must be demonstrated through rigorous, repeatable physical experiments. This motto established empirical observation as the core principle of modern British medical science.

[1.4] Philosophical Transactions and Empirical Peer Review (1665)

"How did scientific journals speed up the exchange of medical knowledge?"

In 1665, Royal Society secretary Henry Oldenburg established the groundbreaking journal **Philosophical Transactions (1665)**, creating the world's first continuous scientific periodical. Oldenburg established a formal network of international scientific correspondence, allowing physicians, chemists, and natural philosophers across Europe to communicate findings, publish experimental records, and challenge unproven theories rapidly in print. This publication revolutionized how academic medical discoveries were evaluated and preserved.

Crucially, the journal pioneered the modern mechanism of **scientific paper peer-sharing**, submitting submitted papers to independent expert committees for critical evaluation before publication. The journal fostered lively **empirical laboratory debates**, recording investigations into blood transfusions, respiration, and microscopic tissue structures. This transparent communication model ensured medical discoveries were tested collectively, preventing fraudulent claims and accelerating genuine scientific progress. It established a permanent archive for European scientific medicine.

[1.5] Early Microscopes and Microscopic Anatomy (1665–1680s)

"Why did 17th-century microscopes fail to transform medical treatment?"

In the seventeenth century, technical advances in lens grinding produced the first **compound light microscopes**, expanding human observation beyond natural vision. In 1665, English natural philosopher **Robert Hooke Micrographia** published his lavishly illustrated masterpiece, presenting breathtaking magnified engravings of fleas, lice, and razor edges, while coining the biological term "cell" after examining slivers of cork tree bark.

In the 1670s, Dutch draper Antony van Leeuwenhoek perfected powerful single-lens microscopes, discovering microscopic bacteria and protozoa in dental scrapings and pond water, famously christened **Leeuwenhoek 'animalcules'**. However, because seventeenth-century glass lenses suffered from chromatic blurring, physicians could not yet link these microscopic organisms to disease, leaving miasma and humoral theories dominant for another two centuries, rendering microscopy a scientific curiosity rather than an immediate clinical tool.

ENQUIRY 02 • PART I

Thomas Sydenham & The Art of Bedside Observation (1676)

KEY ENQUIRY QUESTION *How did Thomas Sydenham's focus on direct observation challenge traditional medical practice?*

[2.1] Thomas Sydenham: The Art of Bedside Observation

> *"Why did Sydenham believe bedside observation surpassed textbook learning?"*

During the mid-seventeenth century, London physician Dr **Thomas Sydenham bedside** pioneer revolutionized clinical medical practice. Nicknamed the "English Hippocrates", Sydenham broke with orthodox university traditions. While academic physicians spent their careers studying classical Galenic texts and drawing astrological charts in libraries, Sydenham insisted that true medical knowledge could only be gained directly at the patient's bedside. He argued that healing required hands-on experience rather than ancient library learning.

Sydenham pioneered **close clinical observation**, spending hours monitoring the patient's physical pulse, skin tone, breathing, and fever patterns. He recorded **detailed symptom logging** across hundreds of clinical cases, meticulously tracking how different illnesses progressed over time. Sydenham taught that doctors should observe natural bodily healing processes without intervening unnecessarily with dangerous purges, fundamentally shifting English medical practice toward rational clinical empircism.

[2.2] Disease Classification and Observationes Medicae (1676)

> *"How did classifying diseases challenge traditional individual humoral theory?"*

Sydenham's greatest conceptual breakthrough was treating diseases as distinct, uniform biological species rather than unique personal humoral imbalances. Applying principles borrowed from **medical botany taxonomy**, he argued that illnesses could be classified into distinct families, genera, and species just like plants. He proved that different patients suffering from the same disease shared identical core physical symptoms. This destroyed the ancient belief that every patient possessed a wholly unique individual illness.

This revolutionary method enabled accurate **differential disease diagnosis**, allowing Sydenham to distinguish measles from scarlet fever for the first time. In his landmark treatise **Observationes Medicae (1676)**, Sydenham published comprehensive diagnostic descriptions of acute epidemics, including smallpox, dysentery, and gout. His textbook became the standard clinical guide studied by European medical students for over two centuries, founding modern clinical pathology.

ENQUIRY 02 • PART II

TURNING POINTS & HISTORICAL CONSEQUENCES

Sydenham & The Concept of Disease Species

DECISIVE HISTORICAL MECHANISM Sydenham moved medicine away from individual humoural imbalance toward specific, identifiable disease entities.

[2.3] Empirical Therapeutics: Cinchona Bark for Malaria

"Why was the success of Peruvian bark a mortal blow to humoral balance?"

In therapeutics, Sydenham championed empirical remedies proven by practical bedside results rather than ancient humoral deductions. When treating malaria, known as ague in seventeenth-century England, he popularized imported **cinchona bark (quinine)** from South America. While traditional Galenic doctors rejected the foreign bark because it fit no humoral category, Sydenham observed that it consistently and miraculously broke dangerous recurring fevers.

By proving cinchona provided **specific cures ague**, Sydenham dealt a devastating blow to traditional humoral medicine, decisively **challenging humour balance** theories that insisted all disease required general bleeding or purging. He also developed a medicinal liquid tincture of opium dissolved in sherry called laudanum, providing reliable pain relief and establishing the principle that specific diseases required targeted chemical cures.

[2.4] The Cool Therapy for Smallpox Fevers

"Why did Sydenham's cool treatment drastically cut smallpox deaths?"

Sydenham's observational methodology led him to challenge dangerous traditional orthodoxies, particularly when managing virulent smallpox epidemics. The standard treatment prescribed by Galenic doctors was the "sweating method", trapping feverish patients in sealed rooms with shuttered windows, piling heavy blankets on beds, and administering boiling cordials to sweat out evil humours, which frequently caused fatal heatstroke in delirious patients.

Sydenham revolutionized care by pioneering his famous **cool therapy treatment**. He ordered **fresh bedroom ventilation**, threw open bedroom windows, removed thick quilts, and kept patients comfortably cool while **avoiding heavy blankets** and serving light broths and barley water. This humane, observational regimen dramatically lowered smallpox mortality, proving that gentle supportive nursing produced far better clinical outcomes than aggressive medical intervention.

[2.5] Conservative Resistance and the Limits of Sydenham's Impact

"Why did medical treatments remain mostly medieval despite Sydenham?"

Despite his clinical brilliance, Sydenham's modern ideas provoked fierce **conservative medical opposition** from London's medical establishment. Elite Fellows of the Royal College of Physicians denounced him as an eccentric troublemaker for ridiculing book-learning and promoting foreign bark remedies. Sydenham was repeatedly denied senior fellowship in the college, and his practical bedside teaching was resisted by conservative university professors. They resented his rejection of ancient philosophical authorities.

Furthermore, Sydenham himself remained constrained by seventeenth-century limitations. Lacking any concept of microscopic bacteria, he still attributed epidemics to atmospheric miasma and practiced **continued purging reliance** when managing non-feverish conditions. Consequently, a very **slow professional shift** occurred across Britain, meaning that while Sydenham transformed diagnostic recording, the majority of ordinary patients continued receiving traditional bloodletting and herbal purges for another century.

ENQUIRY 03 • PART I

PEARSON EDEXCEL GCSE (9–1) HISTORY • PAPER 1 (1HI0/11)

Andreas Vesalius & The Anatomical Revolution (1543)

KEY ENQUIRY QUESTION *Why did Andreas Vesalius succeed in overturning 1,400 years of Galenic anatomical authority?*

[3.1] Andreas Vesalius and Anatomical Dissection at Padua (1537)

"Why was a professor dissecting human bodies revolutionary in 1537?"

Appointed Professor of Surgery and Anatomy at the University of Padua in 1537, Andreas Vesalius transformed medical education across Europe. Breaking radically with medieval custom, where professors sat high upon chairs reading Latin texts while lowly barber-surgeons butchered corpses below, Vesalius entered the **Padua dissection theatre** to conduct **direct personal dissection** with his own hands before packed student audiences.

Vesalius insisted that medical students must verify every anatomical structure with their own eyes rather than trusting ancient manuscripts. Through meticulous human dissections, Vesalius began **challenging Galen's 300 errors**, realizing that the ancient Roman authority had dissected animals like apes, dogs, and pigs because Roman law forbade dissecting human cadavers, leading to hundreds of false assumptions about human anatomy.

[3.2] De Humani Corporis Fabrica: A Printing Masterpiece (1543)

"How did Vesalius prove Galen had made anatomical mistakes?"

In 1543, at the young age of twenty-eight, Vesalius published his revolutionary anatomical masterpiece, **De Humani Corporis Fabrica** (On the Fabric of the Human Body). Printed in Basel by skilled printer **Johannes Oporinus Basel printing**, the monumental folio volume combined rigorous Renaissance scientific observation with the cutting-edge reprographic technology of movable metal type and woodblock printing.

The Fabrica featured over two hundred stunning anatomical illustrations produced by Renaissance master artists from **Titian studio woodcut plates**. Skeletons and muscle figures were depicted posing dynamically against Italian landscapes, displaying layers of dissected muscle tissue with forensic accuracy. The book set a new global standard for anatomical publication, allowing accurate anatomical knowledge to be mass-produced and studied simultaneously across European universities.

ENQUIRY 03 • PART II

TURNING POINTS & HISTORICAL CONSEQUENCES

The Anatomical Breakthrough: Vesalius vs Galen

DECISIVE HISTORICAL MECHANISM Vesalius destroyed the myth of classical infallibility and established direct dissection as medical training.

[3.3] Correcting Ancient Dogma: Jawbone, Septum and Sternum

"Why was The Fabrica considered a masterpiece of Renaissance science?"

Through direct human dissection, Vesalius systematically corrected Galen's most famous anatomical dogmas. Vesalius proved conclusively that the **human jawbone single bone** was one fused bone, refuting Galen's ancient claim that it comprised two distinct halves like that of a dog. Furthermore, he demonstrated that the human breastbone, or **sternum three segments**, was made of three parts rather than seven as found in monkeys.

Most significantly for physiology, Vesalius searched extensively for the **missing heart septum pores** that Galen claimed allowed blood to pass directly between the left and right ventricles. Finding solid muscular tissue with zero invisible holes, Vesalius frankly admitted that blood could not cross the septum, sowing the critical doubts about cardiovascular flow that paved the way for William Harvey's later circulatory breakthrough.

[3.4] Academic Backlash and Resignation from Padua (1544)

"How did English translations spread Vesalius's ideas to barber-surgeons?"

Vesalius's audacious corrections of ancient authority ignited a **vicious traditional backlash** from conservative university faculties across Europe. Traditional Galenists were outraged that a young Belgian professor dared challenge the revered master of classical medicine. In Paris, his former anatomy professor, Jacobus Sylvius, maintained fierce **Sylvius Galenic loyalty**, dismissing Vesalius as an ungrateful madman and claiming that human anatomy had degenerated since Roman times rather than Galen being mistaken.

Embittered by personal hostility, academic slander, and professional ostracism from conservative colleagues, Vesalius made a dramatic **resignation from Padua** in 1544. In a fit of fury, he burned many of his unpublished scientific research notes and abandoned academic dissection entirely, moving into imperial diplomatic service as royal court physician to Holy Roman Emperor Charles V, never publishing major anatomical works again.

[3.5] English Dissemination: Thomas Geminus and Surgical Training

"Why did Vesalius's anatomical discoveries fail to save sick patients immediately?"

Despite academic resistance, Vesalius's revolutionary anatomy spread rapidly across Europe, exerting an immediate practical influence in Tudor England. In 1545, London engraver Thomas Geminus published **Thomas Geminus Compendiosa**, a plagiarized English edition that copied Vesalius's woodcuts into clear copperplate engravings, translating the Latin anatomical text into everyday English so ordinary craft practitioners could understand it. This made modern anatomical science accessible to English readers.

Geminus's affordable manual became the standard handbook for **English barber-surgeon training**, transforming practical surgical apprenticeship in London. Although knowing correct anatomy could not cure internal infections without antiseptics, Vesalius laid the indispensable **anatomical dissection foundation** for modern medicine. By proving that human anatomy could only be learned through dissection rather than book-learning, he inspired generations of Renaissance medical pioneers.

ENQUIRY 04 • PART I

PEARSON EDEXCEL GCSE (9–1) HISTORY • PAPER 1 (1HI0/11)

William Harvey & The Circulation of the Blood (1628)

KEY ENQUIRY QUESTION *Why did William Harvey's discovery of blood circulation meet fierce medical resistance?*

[4.1] William Harvey and Dissecting Cold-Blooded Animals

> *"Why did Harvey study cold-blooded animals to understand the human heart?"*

English physician William Harvey studied medicine at the renowned University of Padua under anatomist **Girolamo Fabricius valves** discoverer. While Fabricius identified tiny one-way flap valves inside human veins, he mistakenly believed they merely slowed blood down to prevent pooling. Harvey suspected these valves served a mechanical purpose, ensuring **one-way blood flow** toward the central heart rather than outward toward extremities.

To observe heart mechanics directly, Harvey conducted pioneering vivisections on slow-beating **cold frog dissections**, toads, and snakes. Because mammalian hearts beat too rapidly for the naked eye, cold-blooded reptiles allowed Harvey to observe systolic contractions clearly, proving that the heart was not a furnace creating heat, but an active muscular pump expelling blood into arterial pathways with each rhythmic contraction.

[4.2] The Mechanical Pump and Mathematical Proof of Circulation

> *"How did mathematical calculations prove Galen's theory impossible?"*

To prove blood circulated continuously, Harvey applied revolutionary quantitative calculations, providing **circulation mathematical proof**. Measuring cardiac ventricles, he calculated that each beat expelled two ounces of blood. With seventy beats per minute, the heart pumped over five hundred pints of blood hourly—three times an adult's body weight—proving it was physically impossible for food to manufacture such massive quantities continually.

Harvey concluded that the **heart mechanical pump** must continuously circulate the exact same blood throughout the body in an endless closed loop. Through this rigorous mathematical proof, the centuries-old **Galenic liver theory refuted** permanently. Harvey proved that the liver did not continuously manufacture fresh blood consumed as bodily fuel, overturning one of the most foundational dogmas of ancient Greek medicine.

ENQUIRY 04 • PART II

TURNING POINTS & HISTORICAL CONSEQUENCES

Harvey & The Mechanical Circuit of the Blood

DECISIVE HISTORICAL MECHANISM Harvey proved the heart is a muscular pump, though bloodletting continued for over a century due to entrenched tradition.

[4.3] De Motu Cordis (1628) and the Tourniquet Experiment

"Why did Harvey publish his research abroad rather than in England?"

In 1628, Harvey published his revolutionary findings in Frankfurt in his masterpiece **De Motu Cordis (1628)** (On the Motion of the Heart and Blood). To convince skeptical physicians, Harvey devised an ingenious public experiment on living human arms: the **forearm tight-ligature test**. By tying a tight cord around a volunteer's upper arm, he cut off arterial flow, leaving the hand cold and pulse-less.

Loosening the band slightly allowed arterial blood to rush into the arm while blocking venous return, swelling the veins. Harvey then conducted his famous **venous valve demonstration**, pressing a finger along a surface vein to push blood backward. The blood refused to flow past venous valves, visibly proving to spectators that blood flowed exclusively toward the central heart in a unified circulatory system.

[4.4] Professional Skepticism and the Limits of Immediate Impact

"Why was Harvey's discovery fiercely rejected by established doctors?"

Harvey's revolutionary discovery encountered intense **conservative medical opposition** across Europe. Traditionalist physicians mocked him as a "circulator" (a Latin pun meaning wandering quack or charlatan), ridiculing the idea that blood circulated in endless loops. In London, Harvey suffered financially by **losing private clients**, as wealthy patients feared being treated by a doctor who rejected the revered authority of ancient Galen. Many patients deserted his private clinic in disgust.

Academic conservatism ensured **slow university adoption**, with medical schools in Paris and London taking nearly fifty years to teach circulation to trainee doctors. Furthermore, Harvey's discovery had zero immediate impact on bedside therapeutics. Knowing blood circulated did not cure diseases, and because physicians still lacked knowledge of germs or blood groups, doctors continued prescribing bloodletting, ironically draining the very blood Harvey had celebrated.

[4.5] Completing the Circulatory Puzzle: Marcello Malpighi (1661)

"Why was the discovery of capillaries the final proof Harvey was right?"

Although Harvey proved mathematically and experimentally that blood must pass from arteries into veins to return to the heart, his theory possessed one major anatomical gap. Without powerful microscopes, Harvey could not explain how blood physically crossed from arterial terminals into venous vessels, hypothesizing that blood somehow soaked through invisible sponge-like flesh before gathering into veins. This missing connection left his circulatory model vulnerable to conservative critics.

The final physical proof came in 1661, four years after Harvey's death, through Italian anatomist **Marcello Malpighi (1661)**. Utilizing newly developed compound microscopes to examine living frog lungs, Malpighi observed delicate **microscopic capillary vessels** connecting arteries directly to veins. With this discovery, the physical mechanism was confirmed and **circulatory proof complete**, vindicating Harvey's mechanical model and laying the cornerstone of modern cardiovascular physiology.

ENQUIRY 05 • PART I

PEARSON EDEXCEL GCSE (9–1) HISTORY • PAPER 1 (1HI0/11)

Continuity in Treatment & The Great Plague of London (1665)

KEY ENQUIRY QUESTION *Why did the Great Plague of 1665 reveal the severe limitations of Renaissance medical progress?*

[5.1] The Great Plague Strikes London: St Giles and Royal Flight (1665)

> *"Why did the rich flee London while the poor suffered catastrophic mortality?"*

In the spring of 1665, catastrophic bubonic plague returned to England with terrifying ferocity during the **Great Plague (1665)**. Originating in the impoverished suburban parish of **St Giles parish outbreak** outside London walls, the epidemic spread rapidly through overcrowded timber slums where raw sewage and rubbish heaps fostered thousands of black rats carrying plague-infected fleas. Summer heat accelerated the bacterial infection into an uncontrollable urban wildfire.

As weekly mortalities escalated, wealthy elites abandoned the capital. King **Charles II court flight** saw the royal family, parliament, wealthy merchants, and senior clergymen evacuate London for rural estates in Oxford and Salisbury. Over thirty qualified physicians and parish priests fled, abandoning over four hundred thousand poor Londoners to face the epidemic with minimal medical assistance or civic support.

[5.2] Supernatural Fears and Humoral Continuity (1665)

> *"Why did people still use 14th-century remedies in 1665?"*

Despite the scientific breakthroughs of Vesalius, Harvey, and the Royal Society, seventeenth-century explanations of disease demonstrated striking continuity with the Middle Ages. The overwhelming majority of citizens believed the epidemic was caused by the **wrath of God & miasma**, interpreting planetary alignments, comets, and solar eclipses as celestial signs of divine vengeance sent upon a decadent, sinful London society.

Medical therapeutics remained steeped in humoral and sympathetic superstition. Citizens purchased expensive **plague water & pomanders** containing cloves, vinegar, and ambergris to ward off corrupt air, while quack doctors peddled miraculous London Drops. In desperate attempts to draw poison from swollen bodies, healers prescribed **pigeon bubo remedies**, strapping the plucked, warm rumps of live pigeons against blackened sores, which achieved zero curative benefit.

ENQUIRY 05 • PART II

TURNING POINTS & HISTORICAL CONSEQUENCES

The Great Plague of 1665: Municipal Intervention

DECISIVE HISTORICAL MECHANISM Key Contrast: While 1348 response was disorganized, 1665 featured organized state and municipal surveillance.

[5.3] Civic Regulation: Quarantines, Red Crosses and Pest Houses (1665)

"How did local government action in 1665 show greater organization than 1348?"

In response to soaring death tolls, London civic authorities enacted far more aggressive public health controls than in 1348. The Lord Mayor issued strict **Mayor quarantine orders**, mandating that any dwelling containing an infected patient be nailed shut from the outside for forty days, trapping healthy family members inside with the sick under the supervision of appointed armed street watchmen.

Quarantined dwellings featured tragic **red cross marked doors**, accompanied by the desperate inscription: *Lord Have Mercy Upon Us!* Dead bodies were surrendered only at night to prevent panic. Authorities also constructed specialized **parish pest houses** in open fields outside London walls, though these primitive wooden isolation barracks had far too few beds to accommodate the infected.

[5.4] Misguided Prevention: Animal Culls and Tobacco Fires (1665)

"Why was slaughtering cats and dogs a tragic medical mistake?"

Because authorities believed the epidemic was spread by atmospheric miasma and animal fur, they enacted well-intentioned but disastrous preventative policies. Convinced household pets carried infection, the Lord Mayor ordered the **culling dogs & cats**, leading to the slaughter of an estimated two hundred thousand animals, which eliminated the natural predators of black rats and accelerated the flea population. This tragic mistake allowed plague-carrying rats to multiply unhindered across London.

To combat corrupt miasma, town authorities maintained continuous **street barrel fires**, burning coal, pitch, and sulfur day and night at road intersections to purify the heavy summer atmosphere. Citizens also practiced **tobacco smoke warding**, compelling children and adults to smoke clay pipes continuously in schools and streets, believing pungent tobacco fumes acted as a powerful protective disinfectant against airborne contagion.

[5.5] The Dead Carts and the Great Demographic Toll (1665)

"How devastating was the demographic impact of the 1665 epidemic?"

By September 1665, London was engulfed in apocalyptic horror as weekly deaths surpassed eight thousand. Traditional churchyard burials ceased completely, replaced by municipal **London dead carts** that traversed midnight streets accompanied by tolling handbells and the grim cries of drivers shouting: *Bring out your dead!* Unclaimed corpses were tipped into gigantic communal burial trenches at Holywell Mount and Aldgate.

By winter 1665, the epidemic subsided as cold weather killed off fleas, leaving over **100,000 plague deaths** recorded across the capital—nearly a quarter of London's total population. Weekly **Bills of Mortality records** documented the horrifying toll across each parish. The catastrophe demonstrated the complete failure of Renaissance medicine to conquer infectious disease, proving that theoretical anatomical progress had yet to transform bedside therapeutics.

Master Knowledge Retrieval Quiz

The Medical Renaissance in Britain • c1500–c1700 (30 Essential Recall Questions)

Standard Recall Protocol: Partner Quiz or Solo Practice. Test your rapid chronological, factual, and conceptual recall across all 5 enquiries. All questions are cross-referenced directly to paragraph locations in this textbook; zero answers are printed to maintain active retrieval integrity.

Enquiry 01: The New Spirit of Enquiry: Hum...

pp. 2–3 • Ref [1.1]–[1.5]

1. Where did the Black Death first arrive in England in June 1348?
2. What bacterium, unknown to medieval people, actually caused the Black Death?
3. What were the agonizing, swollen lymph glands in the groin and armpits called?
4. How was the pneumonic form of the plague transmitted between humans?
5. Who were the Flagellants who appeared during the Black Death epidemic?

Enquiry 02: Thomas Sydenham & The Art o...

pp. 4–5 • Ref [2.1]–[2.5]

6. How did Johannes Gutenberg's movable metal type printing press (c.1440) directly transform medical knowledge?
7. What is the official Latin motto of the Royal Society, and what is its English translation?
8. What was the name of the world's first peer-reviewed scientific journal, published by the Royal Society in 1665?
9. What was the 'Renaissance Paradox' in medical history?
10. In what year did William Caxton introduce the first movable-type printing press to England at Westminster?

Enquiry 03: Andreas Vesalius & The Anato...

pp. 6–7 • Ref [3.1]–[3.5]

11. What nickname was given to Thomas Sydenham because of his emphasis on bedside observation?
12. Name the 1676 book in which Sydenham set out his methods of clinical observation.
13. What treatment did Sydenham prescribe for malaria fevers, imported from South America?
14. Which two childhood illnesses did Sydenham successfully prove were separate diseases?
15. How did Thomas Sydenham's approach to diagnosing disease differ fundamentally from medieval humoural physicians?

Enquiry 04: William Harvey & The Circulati...

pp. 8–9 • Ref [4.1]–[4.5]

16. In what year did Andreas Vesalius publish De Humani Corporis Fabrica?
17. At which famous Italian university did Vesalius serve as professor of surgery?
18. How many anatomical errors made by Galen did Vesalius identify and correct?
19. State one specific anatomical error of Galen corrected by Vesalius.
20. Which ancient Roman medical authority did Andreas Vesalius directly challenge through his public human dissections?

Enquiry 05: Continuity in Treatment & The ...

pp. 10–11 • Ref [5.1]–[5.5]

21. In what year did William Harvey publish De Motu Cordis?
22. Which ancient theory of blood production did Harvey disprove through mathematical calculations?
23. What tiny blood vessels connecting arteries and veins could Harvey NOT see without a microscope?
24. Why did Harvey's discovery of blood circulation have NO immediate effect on medical treatment?
25. How did William Harvey calculate that the liver could not possibly manufacture new blood from digested food?

Synoptic Synthesis & Historical Causation

pp. 2–11 • Whole-Era Recall

26. **[Synoptic]** Explain the single biggest obstacle to medical progress across this era.
27. **[Synoptic]** Which institution or technological breakthrough triggered the greatest shift in clinical practice?
28. **[Synoptic]** How did contemporary beliefs about the causes of disease influence the treatments administered?
29. **[Synoptic]** Assess the extent of continuity between the start and end of this chronological period.
30. **[Synoptic]** Identify one individual whose contributions laid the groundwork for subsequent breakthroughs.